How Forces Affect Motion

Important Questions with Answers

Class 9 Science | NCERT Chapter 6 | 42 Questions

MCQ · True/False · Fill in the blanks · Short answer · Numericals · Long answer

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Tip: cover the green answer box, try the question first, then check. Compare with your own textbook wording.

A. Multiple Choice Questions

Q1 - Q5

Q1. The SI unit of force is:

(a) joule (b) newton (c) watt (d) pascal

Ans: (b) Newton (N). Force is a vector, so both magnitude and direction must be specified.

Q2. Two forces of 10 N and 6 N act on a block in opposite directions. The net force is:

(a) 16 N (b) 4 N in the direction of the 10 N force (c) 4 N in the direction of the 6 N force (d) zero

Ans: (b) For opposite forces, net force = 10 N − 6 N = 4 N, acting along the larger force.

Q3. The tendency of an object to resist a change in its state of rest or uniform motion is called:

(a) friction (b) inertia (c) weight (d) acceleration

Ans: (b) Inertia. Newton used this idea to frame his first law of motion.

Q4. A net force of 20 N acts on an object of mass 4 kg. Its acceleration is:

(a) 0.2 m s^{-2} (b) 5 m s^{-2} (c) 16 m s^{-2} (d) 80 m s^{-2}

Ans: (b) 5 m s^{-2}. a = F ÷ m = 20 ÷ 4 = 5 m s^{-2}.

Q5. An object is moving with a constant velocity. The net force acting on it is:

(a) in the direction of motion (b) opposite to the motion (c) zero (d) equal to its weight

Ans: (c) Zero. By Newton’s first law, constant velocity means no net force (zero acceleration).

A. Multiple Choice Questions (contd.)

Q6. A rocket lifts off because:

(a) the exhaust gases push it upward (b) the air below pushes it up (c) Earth stops attracting it (d) its weight becomes zero

Ans: (a) The engine expels gas downward; by the third law the gas pushes the rocket upward with an equal and opposite force.

Q7. The weight of a 3 kg object (take g = 10 m s^{-2}) is:

(a) 0.3 N (b) 3 N (c) 13 N (d) 30 N

Ans: (d) 30 N. F = mg = 3 × 10 = 30 N.

Q8. An airbag reduces injury in a collision mainly because it:

(a) increases the time over which the passenger stops (b) increases the passenger's mass (c) increases the speed (d) removes friction

Ans: (a) A longer stopping time gives a smaller acceleration, so by F = ma the force on the passenger is smaller.

B. True or False

Q9 - Q14

Q9. Force is a vector quantity.

Ans: True. It has both magnitude and direction, and its SI unit is the newton.

Q10. Balanced forces are equal in magnitude and opposite in direction.

Ans: True. Their net force is zero, so they do not change the motion of the object.

Q11. An object moving with constant velocity must have a net force acting on it.

Ans: False. A net force is needed only to change the velocity; with zero net force, the velocity stays constant.

Q12. The force of friction helps us walk forward.

Ans: True. The foot pushes the ground backward and the ground’s friction pushes the person forward.

Q13. For a given force, the acceleration of an object is directly proportional to its mass.

Ans: False. For a given force, acceleration is inversely proportional to mass (a = F ÷ m).

Q14. The action and reaction forces in Newton’s third law act on the same object and cancel each other.

Ans: False. They act on two different objects, so they do not cancel.

B. True or False (contd.)

Q15. The acceleration due to gravity g does not depend on the mass of the falling object.

Ans: True. Near the Earth’s surface g is about 9.8 m s^{-2} for all objects.

Q16. The smaller the force of friction, the larger the distance a sliding object travels before stopping.

Ans: True. Less friction slows the object more gradually (coins on wood, laminate and marble).

Q17. In a system of two boxes joined by a string, the tension is an external force.

Ans: False. The tension is an internal force of the system; only external forces such as the pull F decide its acceleration.

Q18. On a perfectly smooth horizontal surface, a moving object would continue to move with constant velocity.

Ans: True. With no friction there is no net force, so the object keeps moving (Galileo’s thought experiment).

C. Fill in the Blanks

Q19. Force is a _____ quantity, as it has both magnitude and direction.

Ans: vector.

Q20. According to Newton’s second law, F = _____.

Ans: m × a (mass × acceleration).

Q21. 1 newton = 1 _____ $m\ s^{-2}$.

Ans: kg (1 N = 1 kg m s^{-2}).

Q22. The forces in Newton’s third law act on two _____ objects.

Ans: different.

Q23. The force of _____ acts between surfaces in contact and opposes the motion.

Ans: friction.

Q24. For two boxes of masses m_1 and m_2 pulled by a force F, the acceleration of the system is a = _____.

Ans: F ÷ (m_1 + m_2).

D. Short Answer Questions

Q25 - Q29

Q25. Define force. Give its SI unit and name the instrument used to measure it.

Ans: Force is a push or pull that can change the state of rest or motion, the speed, the direction or the shape of an object. It is a vector. The SI unit is the newton (N), measured with a spring balance.

Q26. Differentiate between balanced and unbalanced forces.

Ans: Balanced forces are equal in magnitude and opposite in direction, so the net force is zero and the motion does not change. Unbalanced forces give a non-zero net force, which changes the velocity of the object.

Q27. You push a heavy box gently but it does not move. Why?

Ans: The force of friction between the box and the floor is equal and opposite to your push. The forces are balanced, so the net force is zero. The box moves only when the applied force is larger than friction.

Q28. State Newton’s first law of motion. What is inertia?

Ans: An object at rest remains at rest, and an object in motion continues to move with constant velocity, unless a net force acts on it. Inertia is the tendency of an object to resist a change in its state of rest or uniform motion.

Q29. State Newton’s second law and write its mathematical form.

Ans: When a net force acts on an object, it accelerates in the direction of the net force. The acceleration is proportional to the net force and inversely proportional to the mass: a = F ÷ m, or F = ma.

D. Short Answer Questions (contd.)

Q30. Define 1 newton.

Ans: One newton is the force that produces an acceleration of 1 m s^{-2} in an object of mass 1 kg: 1 N = 1 kg m s^{-2}.

Q31. Why does a fielder pull the hands backward while catching a fast cricket ball?

Ans: Pulling the hands back increases the time in which the ball's velocity falls to zero. The acceleration is smaller, so by F = ma the force applied is smaller, and injury to the fielder is reduced.

Q32. Why is it difficult to walk on a wet polished floor?

Ans: Friction between the foot and the floor is very small there. Walking depends on the forward friction force given by the ground, so the foot slips backward and we may fall.

Q33. State Newton's third law. Why do the action and reaction forces not cancel each other?

Ans: Whenever one object exerts a force on a second object, the second exerts an equal and opposite force on the first. The two forces act on two different objects, so they cannot balance each other.

Q34. Why does the Earth not appear to move towards a falling fruit although both exert equal forces on each other?

Ans: The forces are equal, but the mass of the Earth is huge compared with the fruit. Since a = F ÷ m, the Earth's acceleration is extremely small and cannot be noticed.

E. Numericals

Q35. A force of 20 N acts on a 5 kg block at rest on a smooth horizontal surface. Find its acceleration, its velocity after 3 s and the displacement in 3 s.

Ans: Acceleration $a = F \div m = 20 \div 5 = 4$ m s^{-2}.
Velocity $v = u + at = 0 + 4 \times 3 = 12$ m s^{-1}.
Displacement $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 4 \times 3^2 = 18$ m.

Q36. A car of mass 1000 kg moving at 20 m s^{-1} is brought to rest in 5 s. Find the force applied by the brakes.

Ans: Acceleration $a = (v - u) \div t = (0 - 20) \div 5 = -4$ m s^{-2}.
Force $F = ma = 1000 \times (-4) = -4000$ N.
The force is 4000 N, opposite to the direction of motion.

Q37. Two boxes of masses 3 kg (Box 1) and 2 kg (Box 2) are joined by a string on a frictionless surface. Box 1 is pulled by a force of 20 N. Find the acceleration of the system and the tension in the string.

Ans: Acceleration of the system $a = F \div (m_1 + m_2) = 20 \div (3 + 2) = 4$ m s^{-2}.
The tension is the only force pulling Box 2: $T = m_2 \times a = 2 \times 4 = 8$ N.
Check for Box 1: $F - T = 20 - 8 = 12$ N $= 3 \times 4$. ✓

F. Long Answer Questions

Q38 - Q40

Q38. Explain friction with the help of the activity using a stack of coins and a spring balance. What conclusion do you draw?

Ans: A stack of coins is released by a stretched rubber band on different horizontal surfaces (wood, laminate, marble), the band stretched equally each time.
• The coins slow down and stop because friction acts opposite to their motion once the band loses contact.
• They travel farther on smoother surfaces, so friction is smaller there.
• Pulling a block with a spring balance on each surface until it just moves gives the friction: a smaller reading means smaller friction.
Conclusion: when friction is smaller, the velocity decreases more slowly and the object travels a larger distance. With zero friction, the object would keep moving with constant velocity (Galileo's thought experiment).

Q39. State Newton's first law. Explain it with examples, and draw (describe) the position-time and velocity-time graphs for an object on which no net force acts.

Ans: Law: an object at rest remains at rest, and an object in motion continues with constant velocity, unless a net force acts on it. If the net force is zero, the acceleration is zero.
Example: a box pushed with a force equal to friction has zero net force, so it keeps moving with constant velocity.
Graphs: at rest, the position-time graph is a horizontal line and the velocity-time graph lies on the time axis ($v = 0$). At constant velocity, the position-time graph is a straight inclined line and the velocity-time graph is a horizontal line.
A net force is needed only to start, stop or change the velocity of an object, not to keep it moving with constant velocity.

Q40. Describe how the relation $F = ma$ is established using the cart and pulley activity. Define the newton.

Ans: A cart is pulled by a thread over a pulley with a paper cup of weights, which gives a constant force. The time T taken to cover the same distance s from rest is measured; since $s = \frac{1}{2}aT^2$, $a_1 \div a_2 = T_2^2 \div T_1^2$.
• Doubling the weight in the cup (more force, same cart mass) gives a larger acceleration: a is proportional to F.
• Doubling the mass of the cart (same force) gives a smaller acceleration: a is inversely proportional to m.
Together: $a = F \div m$, or $F = ma$, with the acceleration along the net force. One newton is the force that gives an acceleration of 1 m s^{-2} to a 1 kg mass: 1 N = 1 kg m s^{-2}.

F. Long Answer Questions (contd.)

Q41. State Newton’s third law. Explain with four everyday examples why the two forces do not cancel each other.

Ans: Law: whenever one object exerts a force on a second object, the second object simultaneously exerts an equal and opposite force on the first.
• Walking: the foot pushes the ground backward; the ground’s friction pushes the person forward.
• Rowing a canoe: the paddle pushes water backward; the water pushes the paddle and canoe forward.
• Rocket (or balloon): the engine expels gas downward; the gas pushes the rocket upward.
• Climbing a tree: the legs push the trunk down; friction pushes the person up.
The two forces always act on two different objects, so they cannot balance each other. Equal forces can still give different accelerations because $a = F \div m$ (gun and bullet: 0.4 m s^{-2} and 20 m s^{-2}).

Q42. Explain how Newton’s laws are applied to a system of two connected boxes. Derive the acceleration of the system.

Ans: Two boxes of masses m_1 and m_2 lie on a frictionless surface, joined by a string. A force F pulls Box 1; the string pulls Box 2 with tension T, and by the third law Box 2 pulls Box 1 back with tension T.
Treat both boxes and the string as one system. Tension T is an internal force and need not be considered; only external forces matter. Here the external force along the surface is F (weight and normal force balance each other).
By Newton’s second law, $a = F \div (\text{total mass}) = F \div (m_1 + m_2)$, the same as a single object of mass $m_1 + m_2$. Treating connected objects as a system simplifies the analysis.